

SCHOOL MANAGEMENT SYSTEM — PROJECT SPECIFICATION DOCUMENT

Version: 1.3

Updated: With School Owner Highest-Level Access

1.0 PROJECT OVERVIEW

The School Management System (SMS) is a web-based platform designed to automate, streamline, and digitize the operations of a primary/secondary school.

The system will support academic management, student data management, parent communication, result processing, attendance, payments, and dashboards for each user type.

2.0 PROJECT GOALS

- Digitize school operations
- Provide real-time insights
- Reduce errors in result processing
- Enable online fee payments
- Improve communication
- Create a scalable platform

3.0 SYSTEM USERS & ROLE HIERARCHY

1. School Owner (Super Admin — Highest Authority)
2. Principal (Admin)
3. HOD / Vice Principal (Sub-Admin)
4. Teachers
5. Parents
6. Students

4.0 ROLE PERMISSIONS

4.1 SCHOOL OWNER (SUPER ADMIN)

Has full control over all modules, can create or disable principal/teacher accounts, configure fee structure, configure school settings, view sensitive financial reports, override principal decisions, and view audit logs.

4.2 PRINCIPAL (ADMIN)

Manages teachers, students, results approval, attendance reports, payment summaries, announcements, and timetables.

Restrictions:

Cannot modify fee structure, cannot change system-level settings, cannot delete owner, cannot access sensitive profit/loss analytics.

4.3 HOD / VICE PRINCIPAL (SUB-ADMIN)

Manages department teachers, reviews results, views analytics for their department, and moderates behavior records.

4.4 TEACHERS

Marks attendance, uploads results, uploads assignments, views class lists, communicates with parents, manages behavioral records.

4.5 PARENTS

Views child information, results, attendance, pays fees, downloads receipts, chats with teachers, views announcements.

4.6 STUDENTS

Views assignments, downloads notes, views results, accesses timetable, views announcements.

5.0 SYSTEM MODULES

1. Authentication & User Access

2. Student Management

3. Parent Portal

4. Teacher Management

5. Class & Subject Management

6. Attendance Management

7. CA & Grading
8. Report Cards & Results
9. Fees & Payments
10. Notifications & Messaging
11. Events & Calendar
12. Assignments & Submissions
13. Behaviour & Discipline
14. Timetable & Scheduling
15. Library Management
16. Transport Management
17. Hostel Management
18. School Admin Dashboard
19. Document Management
20. Owner Dashboard & Audit Logs

6.0 SYSTEM ARCHITECTURE

Frontend: Next.js

Backend: Next.js API Routes or Node.js service

Database: PostgreSQL / MySQL / MongoDB

Auth: JWT / NextAuth

Deployment: Vercel + Railway/Supabase

7.0 USER FLOWS

Login Flow:

Owner → Owner Dashboard

Principal → Admin Dashboard

Teacher → Teacher Dashboard

Parent → Parent Portal

Student → Student Dashboard

Result Processing:

Teacher → HOD → Principal → Publish → Parent/Student View

Attendance Flow:

Teacher marks → Stored → Admin overview → Parent view

Payment Flow:

Parent pays → Receipt generated → Owner/Admin see updates

8.0 DATABASE STRUCTURE (OVERVIEW)

Tables:

users, roles, students, teachers, parents, classes, subjects, attendance, assessments, results, behaviour_records, payments, announcements, assignments, submissions, audit_logs, school_settings

9.0 SECURITY REQUIREMENTS

- JWT sessions
- Password hashing
- RBAC
- Audit logs
- DB encryption
- Input validation

10.0 PERFORMANCE REQUIREMENTS

- Dashboard load <1s
- Pagination
- Caching
- Mobile-responsive UI

11.0 DEPLOYMENT STRUCTURE

Single Deployment Model:

Frontend + Backend on Vercel

Database on Railway/Supabase

Storage on Cloudinary/Supabase Storage

12.0 ADMIN DASHBOARDS

Owner Dashboard:

Income, balances, total students, teacher overview, logs, analytics

Principal Dashboard:

Attendance overview, pending results, teacher activity, events, performance summary

13.0 MILESTONES & DELIVERY PLAN

UI/UX — 2 weeks

Auth — 1 week

Admin Module — 2 weeks

Student/Teacher Management — 2 weeks

Attendance/Results — 3 weeks

Payments — 1 week

Portals — 2 weeks

QA — 1 week

Deployment — 3 days

14.0 CONCLUSION

This document outlines the complete requirements of the SMS, including the updated role hierarchy where the School Owner is the highest authority.